Chimdiume Azie

AI/ML Developer · B.Sc. Computer Science (2026)

Nigeria (UTC+1) · Open to remote work and relocation

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Profile

Computer Science graduate who builds AI systems end to end, from model training and explainability to the application around them. Built a deep learning system that classifies Alzheimer's stages from brain MRI and explains each prediction, with a full hospital web app, running on a laptop with no GPU. Building Nellow, an AI agent that pairs a local Llama model with Claude and answers with sources from the live web. Also designed and built a live WhatsApp-ordering website for a Nigerian food business. Seeking an entry-level AI/ML or software engineering role.

Projects

NeuroScan AI: Alzheimer's Stage Detection with Explainable AI · Final-year project · github.com/Achimdi/neuroscan-ai

- Trained an EfficientNet-B0 classifier (PyTorch, transfer learning) to sort brain MRI scans into four stages (NonDemented, VeryMildDemented, MildDemented, ModerateDemented) on the OASIS MRI dataset (44,355 training images), using a class-weighted loss to handle imbalance.
- Added Grad-CAM heatmaps that show which regions drive each prediction; for Mild and Moderate cases they concentrated on the hippocampus, medial temporal lobe and ventricles.
- Built a Flask hospital web app with Admin and Doctor roles (JWT authentication, role-based access, one-time registration codes), patient records, MRI upload returning predicted stage, confidence and heatmap, saved result history, and an audit trail for patient transfers (SQLite, SQLAlchemy).
- Trained and ran the whole system on a laptop with no GPU, to suit resource-limited hospitals.
- Results: 98.66% validation accuracy and 77.35% on the held-out test set. I traced the gap to a skewed original split (88.5% NonDemented in the test set against 40.8% in training) and documented stratified splitting and clinical validation as next steps. Research prototype, not clinically validated.

Nellow: AI Agent · In development · github.com/Achimdi/nellow

- Built an agent that searches the live web, reads what it finds and answers with sources; runs on local hardware or in the cloud.
- Designed a two-model hybrid: a Llama model runs locally at no per-request cost for quick, low-reasoning answers, and Claude is the cloud reasoning model for deeper, multi-step thinking.
- Built a web workspace with chat history, projects, a speed/model selector, file attachments, and voice input and spoken output controls. Hosting is planned.

Twzee Foods: Menu and WhatsApp Ordering Website · Designer and developer · twzee.foodss.workers.dev

- Designed and built a mobile-first website for a Nigerian healthy-food business: visitors browse the full menu, choose sizes and extras, and send a ready-written order to the business on WhatsApp.
- Structured the site so every price lives in one file the owner can update without touching the code.
- Built with plain HTML, CSS and JavaScript, version-controlled with Git and GitHub, and hosted on Cloudflare.

Skills

- **Machine learning and AI:** PyTorch, transfer learning, CNNs, Grad-CAM and explainable AI, scikit-learn, OpenCV, NumPy, Pandas, model evaluation (precision, recall, F1, confusion matrix, ROC)
- **LLM applications:** AI agents, hybrid local and cloud model design (Llama, Claude), live web search with sourced answers
- **Web and backend:** HTML, CSS, JavaScript, web design (mobile-first), Flask, REST APIs, JWT authentication, SQLite and SQLAlchemy
- **Mobile:** Flutter (working knowledge)
- **Tools:** Python, Git and GitHub, VS Code, Cloudflare

Education

B.Sc. Computer Science, Godfrey Okoye University, 2026

Coursework: Expert Systems, Compiler Construction, Computer Networks and Communication, Human Computer Interaction, Systems Analysis and Design